RNA-Seq Literature Familiarization and Data Characterization

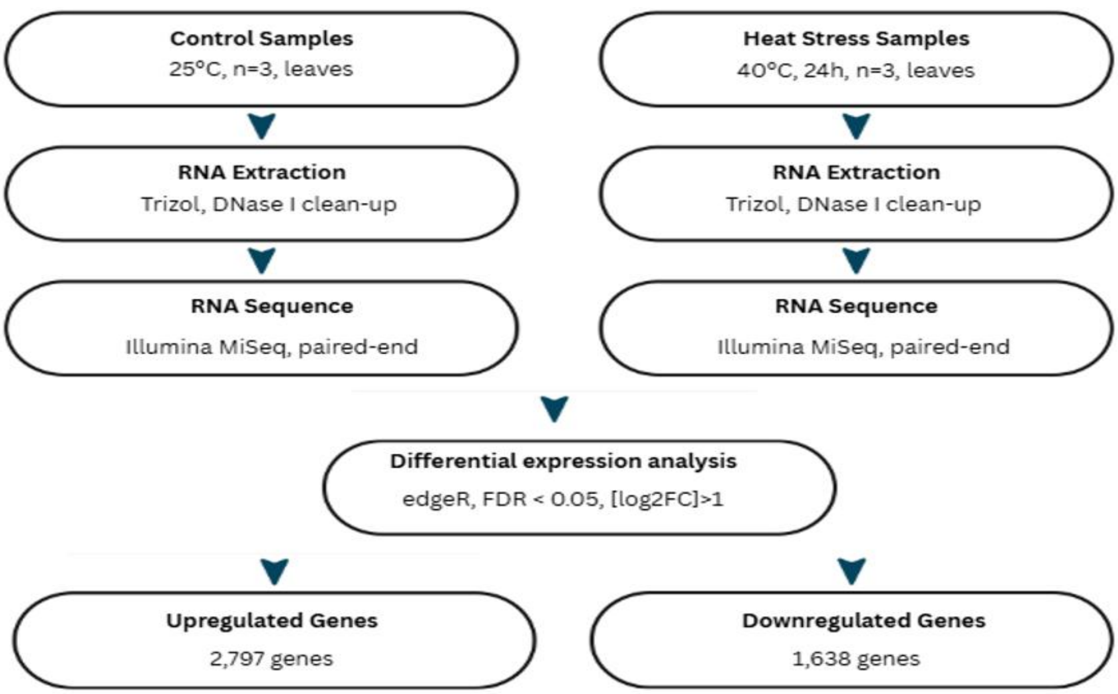
Student: Jasmin Grace T. Bullanday   Group #: 1   Assigned Topic: Heat Stress

Part 3. Paper Information (Group Answer)

Item	**Answer**
Title of the paper	Transcriptomic profiling of *Paulownia elongata* in response to heat stress
Authors	Neerja Katiyar, Niveditha Ramadoss, Dinesh Gupta, Suman B. Pakala, Kerry Cooper, Chhandak Basu
Year	Year 2021
Journal	www.elsevier.com/locate/plantgene
Organism	Paulownia elongata plant
Tissue / organ / cell type	Tissue: leaves Organ: leaves (from 3-year-old plants) Cell type: mixed leaf cells (not specified further; bulk RNA-seq)
Biological question	Which genes and biological pathways in *Paulownia elongata* change their expression in response to heat stress, helping explain heat tolerance? The study identified 4,435 differentially expressed genes.
Control condition	Seedlings grown under normal temperature conditions: 25 °C.
Treatment or stress condition	Heat-stressed seedlings exposed to 40 °C for 24 hours.
Number of biological replicates	Three biological replicates per condition
Sequencing platform	Sanger/Illumina MiSeq
Single-end or paired-end sequencing	Paired-end sequencing
Read length, if reported	35-76 bp (paired-end)
RNA-seq repository	GEO / SRA (Series GSE119074; Study SRP158908)
BioProject / Study accession	PRJNA488054 (BioProject)

Item	Answer
	SRP158908 (SRA Study) GSE119074 (GEO Series)
Run accession numbers	SRR7758328 (Heat_treated_sample1 / SRX4614115) Corresponding runs for the other five samples (Heat_treated_sample2/3 and Control_sample1/2/3) under the same SRP158908
Reference genome or transcriptome used by the authors	The authors used the available *Paulownia elongata* transcriptome/reference sequences for read alignment and annotation, rather than a complete chromosome-level reference genome.
Main RNA-seq software or tools reported by the authors	**Trimmomatic** for read-quality trimming; **TopHat2** for mapping/alignment; **Cufflinks/Cuffdiff** for transcript assembly and differential-expression analysis; and **Blast2GO** for Gene Ontology functional annotation.

Part 4. Experimental Design Diagram



1. What was changed or manipulated?

- Temperature / heat stress (plants exposed to 40 °C for 24 hours)

2. What was the control?

- Plants grown under normal temperature conditions (25 °C).

3. What was measured?

- Gene expression / the transcriptome (RNA-seq), specifically changes in transcript abundance and the identification of 4,435 differentially expressed genes.

4. How many biological replicates were used?

- Three biological replicates per condition (3 heat-treated + 3 control).

5. What tissue or cell type was sequenced?

- Leaves (from 3-year-old Paulownia elongata plants).

Part 6. My Assigned RNA-Seq Sample

Item	**Answer**
Run accession number	SRR7758328
Condition (control/treatment)	Heat_treated_sample1
Biological replicate number, if known	1
Single-end or paired-end	Paired-end
Approximate file size, if available	317.4 mb

Part 8. FASTQ File Examination

1. How is a FASTQ file different from the FASTA genome file used in the previous assignments?

- A FASTA file contains only nucleotide (or amino acid) sequences with simple header lines (starting with >). A FASTQ file contains the raw sequencing reads plus quality information for every base. It uses a four-line format per read instead of the two-line format of FASTA.

2. What information is present in FASTQ that is not present in FASTA?

- Quality scores (Phred scores) for every base
- A second header line (starting with +)
- The actual sequencing quality string that corresponds base-by-base to the sequence

3. Why are quality scores important in RNA-seq?

- Quality scores indicate how reliable each base call is. Low-quality bases can introduce errors that lead to incorrect alignment, false transcript quantification, or spurious differential expression results. Quality scores are used during quality control (e.g., trimming, filtering) to remove or correct low-confidence bases/reads before downstream analysis.

Part 10. My RNA-Seq Dataset Characterization (from FastQC/Galaxy)

Item	**Answer**
RNA-seq run accession	SRR7758328
Condition	Treatment (heat treated)
Single-end or paired-end	Paired-end
Number of reads / sequences	4,956,271 millions read pairs
Read length	35-76 bp
File size	317.4 mb
GC content	45
General per-base sequence quality	Good / Pass (high quality scores, mostly 36–38 across the read; all bases in the green zone)
Presence of adapter content	None
Presence of overrepresented sequences	Not Flagged
Any quality warning observed	Yes – Per base sequence content failed

Part 12. Group Comparison Table

Student	**Accession**	**Condition**	**Reads**	**Read length**	**GC %**	**Main QC observation**
BULLANDAY	SRR7758328	Treatment (heat treated)	4,956,271 million	35-76 bp	45	Per base sequence content failed
ALAMA	SRR7758331	Control	3,147,060 million	35-76 bp	39	R1 failed "Per Base Sequence Content"; mild duplication/length-distribution

Student	Accession	Condition	Reads	Read length	GC %	Main QC observation
						warnings; and trace polyA adapter content
CALIPAS	SRR7758330	Treatment	3,441,734 million	35-76 bp	41	Overall quality:per base sequence content content failed and sequence length distribution warned
FERRER	SRR7758332	Control	1,349,099 million	35-76 bp	41	Per base seq. content FAILED; Seq. Length WARNING

Part 13. Interpretation Questions

1. What biological question was the original RNA-seq study trying to answer?

Which genes and biological pathways in *Paulownia elongata* change expression in response to heat stress (40 °C), helping to explain its heat tolerance? (They identified 4,435 differentially expressed genes.)

2. Why did the authors use RNA-seq instead of only examining the genome?

RNA-seq was used because the genome sequence itself does not change with temperature. Only by measuring RNA levels can researchers see which genes are turned up or down in response to heat.

3. What is the difference between genomic DNA and the RNA molecules measured by RNA-seq?

Genomic DNA is the stable genetic material present in essentially every cell. The RNA molecules quantified by RNA-seq are temporary transcripts whose abundance reflects current gene activity and can change rapidly with environmental conditions.

4. What is a biological replicate and why is it important?

A biological replicate is an independent biological sample (for example, a different plant) that received the identical treatment. Replicates are essential for distinguishing true biological differences from random variation.

5. What is the difference between single-end and paired-end sequencing?

In single-end sequencing only one end of each fragment is read. In paired-end sequencing both ends are read, providing higher mapping accuracy, better detection of splice junctions, and improved resolution of repetitive regions.

6. What is a FASTQ file?

A FASTQ file is a plain-text format that stores raw sequencing reads. Each read occupies four lines: a header, the nucleotide sequence, a separator line, and a corresponding string of quality scores

7. What information does FastQC provide?

FastQC evaluates the technical quality of raw sequencing data. It reports total read number, read length, GC content, per-base quality, adapter content, over-represented sequences, and other diagnostics that indicate whether the data are suitable for downstream analysis.

8. What does a high per-base quality score indicate?

A high per-base quality score means the sequencer assigned that base with high confidence (low probability of error). Scores in the mid-to-high 30s are considered excellent.

9. Why can adapter contamination be a problem?

Adapter sequences are artificial and not part of the biological sample. If retained, they can cause misalignment, reduced mapping rates, and inaccurate quantification of gene expression.

10. Were all RNA-seq samples in your group similar in quality? Explain.

Overall, the samples in the group were of comparable quality. Read lengths and GC contents were similar, and the same mild QC pattern (mainly per-base sequence content bias) appeared across samples. Read counts varied but remained within an acceptable range.

11. Did any sample show a possible quality problem? What was it?

Yes. Multiple samples, including the heat-treated sample SRR7758328, showed a failure in the Per base sequence content module. This is a frequent and generally non-critical feature of RNA-seq data caused by random-hexamer priming bias.

12. What additional steps would be needed before the researchers could compare gene expression between control and treatment samples?

Before differential expression can be performed, the reads must be quality-trimmed and adapter-cleaned, aligned or quasi-mapped to a reference transcriptome, quantified, normalized, and then statistically compared between control and treatment groups (for example with edgeR or DESeq2).

https://github.com/rhealynalama-bio/RNASeq_Group-1_Heat-Stress/tree/5bb905156faed401a70a6d59b26bfabbb03ebe15/BULLANDAY